Computation as Geometric Flow: An Arb-Certified Local Intrinsic ODE on a Quantum-Control Response Fibre

Y. Y. N. Li
Independent Researcher

Abstract

A computational task specified by a fixed finite response may admit many parameter realizations. Yet a response-invisible direction need not be objective-flat: exact preservation of the declared response can coexist with strict decrease of an independent implementation objective. We certify this possibility in a driven-qubit model with projective response $\mathcal{R}_3$ as the retained task and sixth-order projective-loss coefficient L_6 as the implementation objective. Using outward-rounded Arb arithmetic, we validate a local response-fibre chart and its intrinsic normalized projected-gradient ODE. For intrinsic ODE time $0 \leq t \leq 10^{-14}$, the certified solution exists, is unique, preserves $\mathcal{R}_3$ exactly within the declared analytic model, and satisfies $dL_6/dt \leq -0.64195 < 0$. A complementary interval certificate establishes broader-domain regularity and strict descent. This computer-assisted theorem is local; global continuation remains open.

1 Introduction

Conventional input–output descriptions identify a computational task by its declared response, yet quantum-control landscapes commonly admit families of controls with the same or nearly the same objective value [9, 10]. Relative to a fixed response, this residual freedom forms a level set. We ask whether that set can support an independently defined intrinsic dynamics. Because neither Euclidean proximity nor unconstrained ambient optimization guarantees preservation of the response, admissible improvement must be organized intrinsically on the level set.

Quantum-control landscape theory primarily characterizes objective topology, but it also includes observable-preserving homotopies such as D-MORPH, which numerically trace level sets while accommodating secondary design goals [9, 10, 18]. Geometric control theory studies dynamical accessibility under admissible controls [12], while quantum-state geometry supplies differential-geometric structures on spaces of physical states [8]. Validated numerics provides rigorous enclosures for nonlinear equations and ODEs [4, 5]. These strands provide important algorithmic and geometric precedents, but they have not yielded a validated local existence-and-uniqueness theorem for intrinsic dynamics on a nonlinear finite-response fibre.

A distinct physical precedent arises in decoherence-free subspaces and subsystems [16, 17], where protected dynamics was identified within the kernel of a noise-cost operator. However, that line of work addressed environmental decoherence, not the intrinsic geometry of the response fibre itself.

To our knowledge, despite two decades of quantum-control landscape analysis and observable-preserving homotopy methods [9, 10, 18], no rigorous certificate has established, on a nonlinear quantum-control response fibre, the joint existence of an intrinsic dynamics that exactly preserves a prescribed finite response while strictly decreasing an independent implementation objective.

Let $\mathcal{R}_3 : \mathbb{R}^{14} \to \mathbb{R}^8$ denote the declared response and let L_6 be an independent implementation objective. For a prescribed response value c, the admissible implementations form the response fibre

$$\mathcal{F}_c = \mathcal{R}_3^{-1}(c).$$

The nontrivial case occurs when a response-invisible direction is not objective-flat:

$$v \in \ker D\mathcal{R}_3(\theta), \qquad DL_6(\theta)[v] < 0.$$

Equivalently, $\ker D\mathcal{R}_3(\theta)$ need not be contained in $\ker DL_6(\theta)$. We seek a well-defined intrinsic dynamics

$$\theta(0) \longrightarrow \theta(t), \qquad \mathcal{R}_3(\theta(t)) = c, \qquad \frac{dL_6}{dt} < 0.$$

This does not challenge the conventional input–output notion of computational equivalence; it adds a geometric dynamical structure inside the corresponding equivalence class.

Writing $J = D\mathcal{R}_3$, the formal Euclidean projector onto the response-preserving directions is

$$P(\theta) = I_{14} - J(\theta)^T \big(J(\theta)J(\theta)^T\big)^{-1} J(\theta), \tag{1}$$

and the associated normalized projected-gradient field is

$$X(\theta) = -\frac{P(\theta)\nabla L_6(\theta)}{\|P(\theta)\nabla L_6(\theta)\|}. \tag{2}$$

This is the unit-speed negative Riemannian-gradient field of $L_6|_{\mathcal{F}_c}$ with respect to the metric induced from $\mathbb{R}^{14}$ [3]. Under full row rank and a nonzero projected gradient, this construction and the regular-level-set theorem are standard. The theorem-bearing problem is to certify this field quantitatively on a nonlinear projective-response fibre: response rank, a unique implicit fibre chart, positive definiteness of the pullback metric, uniform nonstationarity, and a self-mapping contractive Picard operator yielding a unique exact response-preserving solution with strict objective descent.

The key methodological insight is the tangent–normal coordinate separation of Section 5, which avoids the severe interval wrapping that destroys ambient-tube approaches (Appendix B.2). Combined with outward-rounded 192-bit Arb arithmetic [14], this coordinate separation makes the required nonlinear enclosures sufficiently sharp to close these certification gates on a six-dimensional domain.

For a frozen fourteen-phase driven-qubit model, the full-row-rank response Jacobian at one certified atlas point determines a six-dimensional local response fibre. On a rigorously enclosed tangent–normal chart, we certify a unique intrinsic normalized projected-gradient solution for $0 \le t \le 10^{-14}$ satisfying

$$\mathcal{R}_3(\theta(t)) = \mathcal{R}_3(\theta(0)), \qquad \frac{dL_6}{dt} \le -0.64195 < 0.$$

A floating-point trajectory cannot establish exact response invariance. The proof follows the computer-assisted architecture of Lanford [1, 5], reducing the analytic statement to fail-closed interval gates recorded in Section 4.

The contributions are therefore threefold: a projective-response formulation separating the retained task $\mathcal{R}_3$ from the higher-order objective L_6; an Arb-certified tangent–normal construction supporting a unique local solution of the intrinsic response-preserving ODE with strict L_6 descent; and a complementary broad-domain certificate establishing regularity and strict atlas descent. Here "exact preservation" refers only to the declared finite response $\mathcal{R}_3$ within the frozen analytic model, not to all higher-order coefficients, the full physical output, or hardware behaviour.

The paper establishes a local result, not complete-child or complete-atlas continuation, arbitrary endpoint connection, long-time existence, or a global response-fibre flow. The immediate open problem is to chain adjacent certified microsteps across a complete child interval while preserving the response-fibre and contraction certificates. The subsequent continuation problem is to pass consistently through chart overlaps, with the longer-term goal of constructing a certified global response-fibre flow. Section 13 states these problems precisely.

The following sections define the model and certification architecture, prove and compare the two results, and record reproducibility and the local-to-global boundary.

2 Driven-qubit model and projective response

2.1 Fourteen-phase pulse product

The pulse-product model follows the standard finite-dimensional quantum-control setting [11]. Let

$$\theta = (\theta_1, \ldots, \theta_{14}) \in \mathbb{R}^{14} \tag{3}$$

be the phase controls. In units with the nominal transverse amplitude $\Omega = 1$, a segment subject to common detuning δ is generated by

$$H_j(\delta) = \begin{pmatrix} \delta & e^{-i\theta_j} \\ e^{i\theta_j} & -\delta \end{pmatrix}. \tag{4}$$

All fourteen segments have the declared decimal duration $\tau = 0.62$. Writing $r(\delta) = \sqrt{1+\delta^2}$, the exact segment propagator is

$$U_j(\delta) = \cos\Bigl(\frac{\tau r}{2}\Bigr) I \;-\; i\,\frac{\sin(\tau r/2)}{r}\, H_j(\delta). \tag{5}$$

The total state from $|0\rangle$ is

$$|\psi(\delta;\theta)\rangle = U_{14}(\delta)\cdots U_1(\delta)\,|0\rangle. \tag{6}$$

In the proof code, the trigonometric functions of the declared decimals are evaluated directly by Arb; binary64 trigonometric values do not enter the theorem-bearing state propagation.

2.2 Projective coordinate and symmetric loss

A fixed reference phase vector θ_{ref} determines the normalized nominal target $|t\rangle$. Its algebraic orthogonal complement is denoted $|t^\perp\rangle$ and is given explicitly in Appendix A. On the chart where the denominator does not vanish, define the projective coordinate

$$z(\delta;\theta) = \frac{\langle t^\perp|\psi(\delta;\theta)\rangle}{\langle t|\psi(\delta;\theta)\rangle} = \sum_{k\ge 0} a_k(\theta)\,\delta^k. \tag{7}$$

The implementation simultaneously propagates the reflected analytic extension $\bar z$ so that, on the real detuning axis, $q = z\bar z = |z|^2$. The symmetric projective loss is

$$L(\delta;\theta) = \frac{q(\delta;\theta)}{1+q(\delta;\theta)} = \sum_{k\ge 0} L_k(\theta)\,\delta^k. \tag{8}$$

We use L_6 as the descent potential.

Definition 1 (Third-order projective response)**.** The response map is

$$\mathcal{R}_3(\theta) = \bigl(\operatorname{Re} a_0, \ldots, \operatorname{Re} a_3, \operatorname{Im} a_0, \ldots, \operatorname{Im} a_3\bigr) \in \mathbb{R}^8. \tag{9}$$

Its Jacobian is $J(\theta) = D\mathcal{R}_3(\theta)$.

Remark 1 (Gauge freedom of the reference phase). The overall phase of $|t\rangle$ is not fixed by the construction. Under $|t\rangle \mapsto e^{i\phi}|t\rangle$ one has $|t^{\perp}\rangle \mapsto e^{-i\phi}|t^{\perp}\rangle$ and hence $a_k \mapsto e^{2i\phi}a_k$. This rotates the pairs $(\operatorname{Re} a_k, \operatorname{Im} a_k)$ by a fixed orthogonal transformation of $\mathbb{R}^8$; consequently q, L, L_6, $\ker D\mathcal{R}_3$ and $\|P\nabla L_6\|$ are unchanged. All theorem-bearing scalars below are therefore independent of this choice.

Definition 2 (Projective-jet filtration)**.** For a prescribed projective jet $c^{(k)} = (c_0, \dots, c_k) \in \mathbb{C}^{k+1}$, define

$$\mathcal{F}_k\big(c^{(k)}\big) = \big\{\theta \in \mathbb{R}^{14} : a_j(\theta) = c_j,\ 0 \le j \le k\big\}. \tag{10}$$

Whenever the prescribed jets are compatible,

$$\mathcal{F}_{k+1}\big(c^{(k+1)}\big) \subseteq \mathcal{F}_k\big(c^{(k)}\big).$$

We call the nested family $\{\mathcal{F}_k\}_{k\ge 0}$ the projective-jet filtration. The present computation constrains the level $k = 3$ and uses L_6 as a higher-order descent potential along its candidate local response fibre.

In the frozen theorem, the retained task and descent objective are declared independently: $\mathcal{R}_3$ specifies the finite response equivalence relation, whereas L_6 is not included among the constrained coefficients and serves as the higher-order implementation objective. The filtration separates the retained response from higher-order implementation objectives: fixing $a_0, \dots, a_3$ defines the response level $\mathcal{F}_3$, while L_6 remains available as a descent potential along it. Exact response invariance is certified only for the intrinsic fibre-chart ODE; the serialized atlas satisfies a distinct, near-tangency statement (Section 10).

Remark 2 (General filtration-level construction). Whenever $\mathcal{R}_k$ has locally constant rank and $P_{\ker D\mathcal{R}_k}\nabla V \ne 0$, the same formula defines a candidate response-preserving projected-gradient field. The present paper certifies only $k = 3$ with $V = L_6$.

Proposition 1 (Regular response levels)**.** *If* $\operatorname{rank} D\mathcal{R}_3(\theta) = 8$*, then the exact level set through θ is locally a six-dimensional embedded submanifold of $\mathbb{R}^{14}$, with tangent space* $\ker D\mathcal{R}_3(\theta)$*.*

This is a direct consequence of the submersion theorem, equivalently the regular-level-set theorem. It is a standard differential-geometric fact and does not by itself assert the certified rank, global connectedness, or exact invariance of any serialized curve.

The coefficient and phase derivatives are produced by exact finite jet recurrences through order six. No finite difference is used in the formal response Jacobian or L_6 gradient.

3 Main certified theorems

The two certified statements are stated here in full, main theorem first. Their hypotheses are constructed in Sections 5–7, and their proofs occupy Sections 8 and 9.

Theorem 1 (Validated intrinsic response-fibre ODE microstep)**.** *For the hash-bound model and atlas input, and for the local fibre chart constructed near child 15 (zero-based indexing) in Section 5, there exists a unique solution of Eq. (18) on*

$$0 \le t \le 10^{-14}$$

inside the certified Picard tube. Along this solution:

(i) the implicit response-fibre graph is uniquely defined;

(ii) the pullback metric $H = W^T W$ is positive definite;

(iii)

$$\mathcal{R}_3(\theta(t)) = \mathcal{R}_3(\theta(0))$$

exactly;

(iv) the intrinsic projected-gradient norm satisfies

$$\|\mathrm{grad}_{\mathcal{F}_3} L_6\| \geq 0.64195291915915;$$

(v) and

$$\frac{dL_6}{dt} \leq -0.64195291915915 < 0.$$

The certified Picard contraction factor is 0.0034878591567594*, and the self-mapping utilization is* 0.00058130985945990*. Here t is the time variable of the intrinsic ODE* (18)*; it is unrelated to the Chebyshev coordinate s and to the affine atlas arclength coordinate ℓ of Theorem 2.*

This is the central result: a response-equivalence class is not merely a static degeneracy but supports a certified intrinsic improvement dynamics. The claim is local—one frozen model, one certified initial point, one short time interval—and does not assert such a flow for all equivalent implementations.

Theorem 2 (Complementary complete-parent-box strict descent)**.** *For the serialized model, corrected atlas, chart, parent interval, and exact sixteen-child cover declared in Section 6, the following statements hold uniformly for every $s \in I$, equivalently for every $\ell \in \ell(I)$, where $\widehat{\gamma} = \gamma \circ s(\cdot)$ is the reparameterization by the affine atlas arclength coordinate ℓ of Eq.* (23)*:*

(i) $J(\gamma(s))$ has full row rank eight;

(ii)

$$\|J(\widehat{\gamma}(\ell))\frac{d\widehat{\gamma}}{d\ell}(\ell)\| \leq 2.3071147819355 \times 10^{-9}; \tag{11}$$

(iii)

$$\|P(\gamma(s))\nabla L_6(\gamma(s))\| \geq 0.65307847481072 > 0; \tag{12}$$

(iv) the oriented pairing satisfies

$$\left(\frac{d\widehat{\gamma}}{d\ell}\right)^T P\nabla L_6 \leq -0.65307847481365 < 0; \tag{13}$$

and

(v)

$$\frac{dL_6(\widehat{\gamma}(\ell))}{d\ell} \leq -0.65307846977005 < 0. \tag{14}$$

The maximum certified right-inverse defect over the cover is $0.12993443884002 < 1$*. This theorem is a complementary broad-domain certificate for the certified atlas curve and does not by itself establish ODE existence.*

Remark 3 (Parameters of Theorem 2)*.* The parameter s labels the serialized local Chebyshev interval, while the certified tangency and loss-derivative bounds are expressed with respect to the affine atlas arclength coordinate ℓ of Eq. (23). Neither parameter is the time variable of the intrinsic ODE in Theorem 1. Statement (iii) is intrinsically independent of the parameterization. Its certified lower bound in the present proof is nevertheless obtained from the parameterized oriented pairing (13) through Cauchy–Schwarz, and therefore uses the accompanying enclosure of $\|d\widehat{\gamma}/d\ell\|$ recorded in Section 7.2.

4 Analytic reduction of the proof

The proof of Theorem 1 is reduced to the following finite list of interval statements, each verified by outward-rounded Arb arithmetic on the declared domains:

(R1) the response Jacobian J has full row rank eight at the chart midpoint;

(R2) the implicit normal graph $b = \psi(a)$ of the fibre equation exists uniquely on the declared complex tangent domain;

(R3) the whitened response Gram matrix is uniformly invertible and the pullback metric $H = W^T W$ is positive definite throughout the tube;

(R4) the intrinsic projected gradient is uniformly separated from zero;

(R5) the intrinsic vector field and its derivatives admit rigorous interval enclosures via Cauchy–Taylor bounds;

(R6) the Picard operator maps the certified tube into itself;

(R7) the Picard operator has Lipschitz constant strictly below one on that tube;

(R8) the tangency identity and the Lyapunov descent identity hold, yielding exact response invariance and strict L_6 decrease.

The computer is not simulating the theorem; it is verifying the finite inequalities on which the analytic proof rests. Sections 5–7 construct the domains and enclosure machinery. Section 8 discharges the microstep obligations (R1)–(R8), with the midpoint rank certificate supplied by Theorem 2(i), proved on the complete parent box in Section 9.

5 Geometry of the response fibre

For the v0.9.3 microstep the chart centre is exact: with subdivision 32 and sixteen children of half-width 2^{-10}, the midpoint of child 15 of the exact cover defined in Eq. (25) is

$$s^* = \frac{31}{1024} = 0.0302734375 \in I, \qquad \theta_0 = \gamma_9(s^*), \tag{15}$$

so θ_0 lies on the certified parent interval $I = [0, 1/32]$ of Section 6; the appeal to Theorem 2(i) in the proof of Theorem 1 is therefore licensed.

At this midpoint, a frozen SVD of $J = D\mathcal{R}_3(\theta_0)$ determines orthonormal matrices

$$T \in \mathbb{R}^{14\times 6}, \qquad N \in \mathbb{R}^{14\times 8},$$

spanning the tangent and response-normal spaces, respectively. We write

$$\theta(a, b) = \theta_0 + Ta + Nb.$$

The same SVD supplies the frozen response whitener $B = \Sigma^{-1} U^T$, where Σ collects the singular values of $J(\theta_0)$ and U is the corresponding left factor. Its use is licensed by the certified invertibility of the composite normal derivative BJN, defined in Section 7.3 and certified in Section 8; since that composite is certified invertible on the chart domain, B is invertible and $F = 0$ is equivalent to $\mathcal{R}_3(\theta(a, b)) = \mathcal{R}_3(\theta_0)$. With this whitening the fibre equation is

$$F(a, b) = B\big(\mathcal{R}_3(\theta_0 + Ta + Nb) - \mathcal{R}_3(\theta_0)\big) = 0.$$

A parametric Krawczyk argument [6] certifies a unique analytic graph $b = \psi(a)$ on the declared complex tangent domain. The fibre chart is

$$\theta(a) = \theta_0 + Ta + N\psi(a), \tag{16}$$

with differential and pulled-back metric

$$W(a) = T + ND\psi(a), \qquad H(a) = W(a)^T W(a). \tag{17}$$

The intrinsic normalized projected-gradient field is

$$\dot{a} = -\frac{H(a)^{-1}W(a)^T\nabla L_6(\theta(a))}{\sqrt{\big(W(a)^T\nabla L_6(\theta(a))\big)^T H(a)^{-1}\big(W(a)^T\nabla L_6(\theta(a))\big)}}. \tag{18}$$

Two proof identities accompany this construction. The tangency identity

$$D\mathcal{R}_3(\theta(a))\, W(a) = 0 \tag{19}$$

expresses exact response-level invariance of the chart, and the Lyapunov identity

$$\frac{dL_6}{dt} = -\sqrt{\big(W^T\nabla L_6\big)^T H^{-1}\big(W^T\nabla L_6\big)} \tag{20}$$

expresses strict descent along any solution of Eq. (18).

6 Serialized atlas and local proof domain

The input is a corrected Chebyshev atlas represented by ten endpoint-locked degree-22 charts

$$\gamma_j : [-1,1] \longrightarrow \mathbb{R}^{14}, \qquad j = 0,\dots,9. \tag{21}$$

The formal theorem concerns only the restriction of chart 9 to the local Chebyshev interval

$$\gamma = \gamma_9\big|_I, \qquad I = [0,\ 0.03125]. \tag{22}$$

Throughout the certified theorem, s denotes this local Chebyshev coordinate; it is not asserted to be physical time or global arclength. Writing $[\ell_-,\ell_+]$ for the declared arclength interval of chart 9, we also use the affine atlas arclength coordinate

$$\ell(s) = \ell_- + \frac{s+1}{2}(\ell_+ - \ell_-), \qquad \frac{d\ell}{ds} = \frac{\ell_+ - \ell_-}{2} > 0, \tag{23}$$

and write

$$\widehat{\gamma} : \ell(I) \longrightarrow \mathbb{R}^{14}, \qquad \widehat{\gamma}(\ell) = \gamma(s(\ell)), \qquad \ell(I) = [\ell(0), \ell(0.03125)], \tag{24}$$

where $s(\cdot)$ is the inverse of the affine map (23) and $\ell(I) \subseteq [\ell_-,\ell_+]$. For chart 9 the serialized atlas declares

$$\ell_- = 0.3601462225733, \qquad \ell_+ = 0.40016246952588885,$$

so that $d\ell/ds = 0.020008123476294425$ and $|\ell(I)| = 6.2525385863420078 \times 10^{-4}$. The distinction is not cosmetic: in the frozen code the quantities `curve_derivative` and `loss_derivative` are derivatives with respect to ℓ, the Chebyshev derivative being converted by the constant factor $2/(\ell_+ - \ell_-)$ of Eq. (23). Accordingly the domain of Theorem 2 is declared in s, whereas its certified rate constants are expressed in ℓ. The atlas's canonical JSON SHA-256 is

`c02acc1c76e0b670793340150d1a875fdc373e0ac7c46d3360a7824b66a3a5ef`.

computed from the UTF-8 encoding of the JSON serialization with sorted keys and the compact separators `(",", ":")`. The atlas was obtained from a floating spectral-collocation reconstruction; that construction is candidate generation, not part of the formal theorem. The present proof treats the serialized coefficients as fixed input and verifies the claimed inequalities independently with outward-rounded arithmetic.

We select chart 9 (zero-based indexing) and subdivision 32. The parent interval I is covered by sixteen exact contiguous children,

$$I_m = \left[\frac{m}{512}, \frac{m+1}{512}\right], \qquad m = 0, \ldots, 15, \tag{25}$$

each with half-width $2^{-10} = 0.0009765625$ in the Chebyshev coordinate s.

This chart is the weakest of the ten charts in the preceding floating screen, in the sense that it has the smallest sampled projected-gradient norm. Selecting it before the formal audit makes the local test conservative, but it does not replace a complete ten-chart proof.

7 Computer-assisted certification machinery

This section supplies the enclosure machinery behind the proof obligations (R1)–(R8) of Section 4. The proof follows the fail-closed principles of interval analysis and validated numerics [4, 5]. Chebyshev representations are used in the standard spectral-approximation sense [7]. All theorem-bearing enclosures are evaluated with Arb midpoint-radius arithmetic [14].

7.1 Dependency-preserving scalar Taylor models

Directly replacing all fourteen phases by independent intervals produces severe wrapping. Instead, on each child we preserve the common scalar dependence $s = s_0 + u$. For an analytic scalar or matrix invariant $F(s)$, the code evaluates F at 64 points on a complex Cauchy circle, extracts Taylor coefficients through order 24, encloses aliasing, and bounds the remaining tail on the real child interval. A second boundary sweep bounds the analytic function on a larger circle; its 64 arcs are complex interval arcs of angular half-width $\pi/64$ that tile the full circle, not 64 sampled boundary points, so the resulting bound M_R is an enclosure of the supremum over the whole boundary. Thus each reported interval contains $F(s)$ for every real s in the child, not only the sampled nodes.

Lemma 1 (Cauchy–Taylor enclosure). *Let F be analytic on an open neighbourhood of the closed disk $\{u : |u| \le R\}$, and suppose that $|F(u)| \le M_R$ there. Let $0 < r < \rho < R$, let $\omega_j = \exp(2\pi i j/N)$, and, for $0 \le n < N$, define the discrete Cauchy coefficients*

$$\widehat{c}_n = \frac{1}{N\rho^n} \sum_{j=0}^{N-1} F(\rho\omega_j)\, \omega_j^{-n}. \tag{26}$$

If c_n are the exact Taylor coefficients of F, then

$$|\widehat{c}_n - c_n| \le M_R\, R^{-n}\, \frac{(\rho/R)^N}{1-(\rho/R)^N}. \tag{27}$$

Moreover, for every $|u| \le r$, the Taylor remainder after degree p satisfies

$$\left| F(u) - \sum_{n=0}^{p} c_n u^n \right| \le M_R\, \frac{(r/R)^{p+1}}{1-r/R}. \tag{28}$$

Proof. The alias identity follows by inserting the Taylor series into the discrete Fourier sum:

$$\widehat{c}_n = \sum_{\ell \geq 0} c_{n+\ell N}\, \rho^{\ell N}.$$

Cauchy's coefficient estimate $|c_m| \leq M_R R^{-m}$ gives Eq. (27). Summing the geometric majorant for degrees above p gives Eq. (28). $\square$

In the frozen audit, $N = 64$, $p = 24$, the real child half-width is $r = 2^{-10}$, the coefficient-extraction radius is $\rho = 2r$, and the analytic bound radius is $R = 3r$. Arb enclosures are used for the discrete coefficients and for M_R; the alias and tail bounds are then added by outward-rounded arithmetic.

The large complex rectangle is first checked to exclude every denominator appearing in the projective coordinate and loss. This provides the analytic domain required by Cauchy's estimate. Chebyshev evaluation and differentiation of γ are also performed in Arb arithmetic.

7.2 Response rank and projected gradient

Set

$$G = JJ^T \in \mathbb{R}^{8\times 8}, \qquad b = J\nabla L_6. \tag{29}$$

At the centre of each child, a floating inverse R of the midpoint Gram matrix and a midpoint solution y_0 are frozen as exact point data. They are not accepted as proofs. The theorem-bearing quantities are

$$E = I_8 - RG, \qquad z = R(b - Gy_0), \tag{30}$$

formed before Cauchy extraction. If

$$q = \|E\|_\infty < 1, \tag{31}$$

then G is nonsingular and the exact solution $y = G^{-1}b$ obeys

$$\|y - y_0\|_\infty \leq \frac{\|z\|_\infty}{1-q}. \tag{32}$$

Throughout this subsection $\dot\gamma := d\widehat{\gamma}/d\ell$ denotes the curve derivative with respect to the affine atlas arclength coordinate. This correction is propagated into the scalar contractions

$$\|P\nabla L_6\|^2 = \|\nabla L_6\|^2 - b^T G^{-1} b, \tag{33}$$

$$\dot\gamma^T P\nabla L_6 = \nabla L_6^T \dot\gamma - (J\dot\gamma)^T G^{-1} b. \tag{34}$$

The left-hand side of Eq. (34) is the oriented pairing of Theorem 2(iv); its sign is invariant under any positive reparameterization. The quantity $\dot\gamma$ is the frozen code's curve derivative, obtained from the Chebyshev derivative by the constant factor $2/(\ell_+ - \ell_-)$ of Eq. (23). The frozen code's loss derivative is likewise $dL_6/d\ell$, not dL_6/ds. The response-tangency residual and loss derivative are enclosed directly:

$$\|J(\widehat{\gamma}(\ell))\frac{d\widehat{\gamma}}{d\ell}(\ell)\|, \qquad \frac{dL_6(\widehat{\gamma}(\ell))}{d\ell} = \nabla L_6(\widehat{\gamma}(\ell))^T \frac{d\widehat{\gamma}}{d\ell}(\ell). \tag{35}$$

Writing $v = d\widehat{\gamma}/d\ell$, the projected-gradient norm also admits the Cauchy–Schwarz lower bound

$$\|P\nabla L_6\| \;\geq\; \frac{|v^T P\nabla L_6|}{\|v\|}.$$

The parent-box audit encloses the curve speed uniformly over the sixteen children as

$$\|v\|^2 \in [\,0.999999999989079,\; 1.00000000000896\,],$$

the endpoints being outward-safe renderings of the minimum and the maximum attained over the cover, and it verifies the fail-closed gate $\inf\|v\|^2 > 0$. This speed enclosure, together with the strictly negative oriented pairing, supplies the certified lower bound of Theorem 2(iii). Equation (33) is retained as an analytic identity; its direct interval propagation is too lossy on the present domain to be the theorem-bearing route.

The frozen KKT alignment witness is diagnostic rather than theorem-bearing and is reported in Appendix B.1.

7.3 Certificate scalars and their gates

Every gate decision quoted in this paper compares one of the following scalars against a threshold fixed before the corresponding run. The definitions are those of the frozen sources; R, C and C_H denote frozen floating midpoint inverses used only as preconditioners.

Scalar	Definition	Gate	Value reported in
right-inverse defect	$\|I_8 - RG\|_\infty$, Eq. (31)	< 0.8	Thm. 2, Sec. 9
graph (Krawczyk) utilization	Krawczyk image radius / normal-graph radius	< 0.95	Sec. 8, step (ii)
normal-derivative defect	$\|I_8 - C\,(BJN)\|_\infty$	< 0.8	Sec. 8, step (iii)
pullback-metric defect	$\|I_6 - C_H H\|_\infty$	< 0.8	Sec. 8, step (iv)
Picard contraction factor	hL, with $L \le dM/(R_c - r_c)$	< 0.5	Thm. 1, Sec. 8
Picard self-mapping utilization	$h \sup\|X\|_\infty / r_c$	< 0.95	Thm. 1, Sec. 8

Here R_c and r_c are the outer complex and inner real microstep radii of Section 8, not the Cauchy radii of Section 7.1. The normal-derivative defect is recorded twice in the frozen certificate, once for the fibre-graph stage and once for the $D\psi$ Neumann solve; the two fields carry the same expression and the same value, and the Neumann solve requires only < 1, so the stricter predeclared threshold < 0.8 governs.

8 Intrinsic ODE microstep certificate

The main certification layer works on the tangent–normal fibre chart of Section 5, constructed near child 15 of the selected Chebyshev chart. It does not reuse the serialized atlas as a trajectory; instead it validates the intrinsic ODE (18) directly. The certificate of Theorem 1 is assembled in three steps: a fibre-graph enclosure discharging (R2)–(R4), resting on the midpoint rank certificate (R1) supplied by Theorem 2(i); a Picard enclosure discharging (R5)–(R7); and the final analytic argument discharging (R8).

8.1 Centered mean-value fibre-graph enclosure

The fibre-graph stage avoids an independent interval subtraction of two nearly equal response values by using the exact identity

$$B\big(\mathcal{R}_3(\theta_0 + Ta) - \mathcal{R}_3(\theta_0)\big) = \int_0^1 B\, D\mathcal{R}_3(\theta_0 + \sigma Ta)\, Ta\, d\sigma. \tag{36}$$

The parameter residual and the normal derivative are enclosed jointly in outward-rounded Arb arithmetic. A parametric Krawczyk inclusion then certifies the unique normal graph $b = \psi(a)$.

8.2 Intrinsic Picard enclosure

The use of interval enclosures to certify ODE existence and uniqueness follows the validated-integration viewpoint of Lohner and Tucker [2, 5]. The microstep audit uses an outer complex tangent radius $R = 2 \times 10^{-11}$ and an inner real Picard radius $r = 10^{-11}$. The selected normal graph radius is $4.636154057451207 \times 10^{-13}$. On the complex microstep box the audit encloses the normalization square $(W^T \nabla L_6)^T H^{-1}(W^T \nabla L_6)$ in the open right half-plane and separates the normalization denominator from zero, so a single analytic branch of the square root is available and the field (18) is analytic there; the audit fails closed if either enclosure is violated. Cauchy's estimate on the complex polydisc then gives a Lipschitz bound for the intrinsic field: with $M = \sup\|X\|_\infty$ taken over the complex polydisc of radius R and $d = 6$ the tangent dimension, the induced sup-norm Lipschitz constant obeys $L \le dM/(R - r)$. The Picard contraction conditions are

$$hL < 1, \qquad \frac{h \sup\|X\|_\infty}{r} < 1.$$

The word *gate* is reserved below for the stricter predeclared audit thresholds. For $h = 10^{-14}$, the certified values are

$$hL \le 0.0034878591567594, \qquad \frac{h \sup\|X\|_\infty}{r} \le 0.00058130985945990.$$

As for Theorem 2, the microstep gates were fixed before the formal run. The frozen v0.9.3 protocol declares them under the following field names:

Protocol field	Gate
`maximum_graph_derivative_defect`	< 0.8
`maximum_graph_utilization`	< 0.95
`maximum_pullback_metric_defect`	< 0.8
`minimum_projected_gradient_norm`	> 0.6
`maximum_L6_derivative`	< -0.55
`maximum_picard_contraction`	< 0.5
`maximum_picard_self_mapping_utilization`	< 0.95

The same protocol records `validated_ODE_claimed_before_audit` = `false`.

Proof of Theorem 1. The logical chain has eight steps, each supported by a certified bound. (i) By Theorem 2(i) at the chart midpoint, $\mathcal{R}_3$ is a submersion there, so the response fibre $\mathcal{R}_3^{-1}(c)$ is locally a six-dimensional regular submanifold. (ii) The centered mean-value enclosure (36) feeds a parametric Krawczyk inclusion whose utilization is $0.023161050514437 < 0.95$; hence the implicit normal graph $b = \psi(a)$ exists uniquely on the declared complex tangent domain. (iii) The normal-derivative defect is 0.00056424395018765, which bounds $D\psi$ and hence the chart differential W. (iv) The pullback metric Neumann defect is $0.00069535968103773 < 1$, so $H = W^T W$ is positive definite throughout the tube and the intrinsic field (18) is well defined. (v) The certified intrinsic gradient lower bound $0.64195291915915 > 0$ separates the normalization denominator from zero on the complex box, so the field is analytic there; Cauchy's estimate on the complex polydisc with outer radius $R = 2 \times 10^{-11}$ then yields the Lipschitz bound $L \le 3.4878591567594 \times 10^{11}$ for the intrinsic field. (vi) With step size $h = 10^{-14}$,

$$hL \le 0.0034878591567594 < 1, \qquad \frac{h \sup\|X\|_\infty}{r} \le 0.00058130985945990 < 1,$$

so the Picard iteration is a contraction mapping the certified tube into itself; hence a unique solution exists on $0 \le t \le 10^{-14}$. (vii) The tangency identity (19) holds identically on the fibre chart, so $\mathcal{R}_3$ is preserved exactly along the solution. (viii) The Lyapunov identity (20), combined with the intrinsic gradient lower bound of step (v), gives the stated strict-descent inequality. $\square$

9 Complementary parent-box certificate

The theorem-bearing gates were fixed before the formal run:

Quantity	Gate
right-inverse defect	< 0.8
projected-gradient norm	> 0.60
response-tangency norm $\|J\, d\widehat{\gamma}/d\ell\|$	$< 10^{-6}$
oriented pairing	< 0
$dL_6/d\ell$	< -0.55
curve-speed square $\inf\|d\widehat{\gamma}/d\ell\|^2$	> 0

Proof of Theorem 2. On each of the sixteen exact child intervals, the dependency-preserving Cauchy and Taylor construction encloses all entries of E, z, the response tangent, the loss derivative, and the scalar contractions in Eqs. (33)–(34). The maximum row-sum defect is bounded above by 0.12993443884002. The Neumann lemma therefore proves that $G = JJ^T$ is invertible on every child, hence rank $J = 8$, and bounds the exact solve correction by Eq. (32). Propagating the solve correction proves the stated negative bound (13) for the oriented pairing. The certified curve-speed enclosure of Section 7.2 and Cauchy–Schwarz then give, on the minimizing child,

$$\|P\nabla L_6\| \;\geq\; \frac{0.65307847481365}{\sqrt{1.00000000000896}} \;\geq\; 0.65307847481072,$$

which proves Theorem 2(iii); the certified bound is the minimum over children of the per-child Cauchy–Schwarz quotient, and the extremal pairing and the extremal speed enclosure are both attained on child 15. Direct evaluation of Eq. (33) is retained as an analytic identity but is not used to obtain this certified lower bound. Direct interval contractions with $d\widehat{\gamma}/d\ell$ give the certified response-tangency and $dL_6/d\ell$ bounds. Every child passes all theorem-bearing gates above, and the children form an exact contiguous cover of I, so the stated bounds hold uniformly on the parent interval. □

Corollary 1 (No local projected-gradient stationary point)**.** *The projected L_6 gradient has no zero on the certified parent interval.*

The certified rank bound verifies the hypothesis of Proposition 1 throughout the complete parent interval: for every $s \in I$, the exact level set through $\gamma(s)$ is locally a six-dimensional embedded submanifold of $\mathbb{R}^{14}$.

Corollary 2 (Strict ordering along the certified interval)**.** *If $s_1 < s_2$ are points of the certified interval, then*

$$L_6(\gamma(s_2)) < L_6(\gamma(s_1)).$$

More quantitatively, with $\ell(\cdot)$ the affine arclength coordinate of Eq. (23),

$$L_6(\gamma(s_2)) - L_6(\gamma(s_1)) \leq -0.65307846977005\,(\ell(s_2) - \ell(s_1)).$$

Since $d\ell/ds = (\ell_+ - \ell_-)/2 > 0$ is a positive constant on the chart, $s_1 < s_2$ implies $\ell(s_1) < \ell(s_2)$, and the ordering statement in s follows immediately from the quantitative bound. Over the complete certified parent interval, $|\ell(I)| = 6.2525385863420078 \times 10^{-4}$ and hence

$$L_6(\gamma(0.03125)) - L_6(\gamma(0)) \leq -4.0833983321464 \times 10^{-4}.$$

Remark 4. The certificate is a uniform near-tangency and descent statement along the serialized atlas, not the identity $J\, d\widehat{\gamma}/d\ell = 0$; exact response invariance by a validated ODE is established separately in Theorem 1, proved in Section 8.

10 Logical comparison of the certificates

The two certificates are logically distinct:

$$\text{broad-domain near-tangent descent} \neq \text{small-domain exact response-preserving ODE.}$$

Certificate	Domain	Parameter	Response	Dynamical strength
Parent box (Thm. 2)	wider interval	domain s, rates ℓ	near-tangency	strict atlas descent
Microstep (Thm. 1)	local fibre tube	t	exact invariance	unique validated solution

The microstep theorem does not identify the serialized Chebyshev atlas with an ODE trajectory; it separately constructs an intrinsic fibre graph and validates its induced ODE. Supporting development diagnostics are recorded in Appendix B.

Proposition 2 (What would complete a chart-level flow theorem)**.** *A validated chart-level projected-gradient-flow statement would require, in addition to Theorems 1 and 2, chaining of adjacent validated microsteps with endpoint inclusion, traversal of a complete child interval, and compatible continuation across chart overlaps. Pointwise or sampled near-alignment of the centre atlas is insufficient. The distinction between a numerical ODE approximation and a validated enclosure of an exact solution is fundamental in numerical ODE analysis and validated integration [13, 2, 5].*

11 Mathematical-physical interpretation

With the proofs complete, the certified results admit three layers of interpretation:

- *Static layer*: fixing the declared response still leaves, locally about each certified point, a six-dimensional parameter freedom;
- *Dynamical layer*: this freedom can carry a genuine intrinsic ODE;
- *Computational layer*: along that ODE, the declared response is exactly invariant while an independent objective decreases strictly.

The certified microstep solution thus separates two notions often conflated in an input–output description: computational identity, represented by the response fibre, and implementation dynamics, represented by motion tangent to that fibre.

There is a limited analogy with gauge freedom: points on a response fibre are indistinguishable only with respect to the declared finite response $\mathcal{R}_3$. Unlike points on a genuine gauge orbit, they may remain distinguishable by the higher-order objective L_6. Hence $D\mathcal{R}_3[X] = 0$ need not imply $DL_6[X] = 0$: a response-invisible direction can still carry response-preserving improvement dynamics. This task-relative analogy does not identify the response fibre with a fundamental gauge orbit.

These statements concern the certified analytic model; their physical realization is not addressed here.

12 Reproducibility and audit trail

The v0.9.3 release archives the source, the exact backend-input archive, the hash-bound protocol, certificate, and report files, and a release verifier. The principal identifiers are listed in Table 1; the complete hash list for the frozen release payload is distributed as `SHA256SUMS.txt` with the release.

Table 1: Principal frozen identifiers (v0.9.3 release). The complete list is in `SHA256SUMS.txt` of the release.

Artifact	SHA-256
corrected atlas (canonical JSON)	`c02acc1c76e0b670793340150d1a875f dc373e0ac7c46d3360a7824b66a3a5ef`
v0.7.4 source (Theorem 2)	`1f71c4918d1cd1d6c45dc0da4a7358e176baac 9116c8f71f4a949a6d657520f8`
v0.7.4 stored certificate file (Theorem 2)	`ed302725c03b66f2bbe7363e901842f07790 bd895b4cf95261ebe3d172d400f9`
v0.7.4 original-run certificate identifier (reported in the reference summary)	`e9082f5a404b9c3f9539492891a9741ec113 542196fd5e5a6997bad83e7715e8`
v0.9.3 source (Theorem 1)	`3be3e07146ff0e505f08bae7bd0ec7f2895955 f2540647fea3278fdba51db79c`
v0.9.3 certificate file (Theorem 1)	`96cd24d34d1b426eef74696c83441510890b 50902ae6cbe60fed3fc741bfbf3c`
backend-input ZIP	`2efd863f5ff26da1067594f068bfe265678e6e bac480574ff0574ccc55f98666`

The stored certificate file and the original-run certificate identifier refer to distinct executions. Their floating preconditioners produce slightly different valid defect bounds across numerical environments. The theorem-bearing enclosures of the exact quantities quoted in Theorem 2 are unchanged; the defect value used in the present text is the value recorded in the shipped release certificate.

The SHA-256 identifiers are part of the certificate boundary: they identify the precise code, inputs, and outputs to which the computer-assisted claims apply. They serve identity binding only—they do not by themselves prove any mathematical statement, and they are not substitutes for the interval proof. A change to any bound input defines a different computational statement and requires a separate certificate.

Reproducers should note the parameter convention of Section 6: the frozen certificate fields `dL6_dell_upper` and `maximum_dL6_dell_upper`, together with the source quantities `curve_derivative` and `loss_derivative`, are all derivatives with respect to the affine atlas arclength coordinate ℓ. Comparing them against a Chebyshev-coordinate derivative d/ds will disagree by the constant factor $d\ell/ds = (\ell_+ - \ell_-)/2$.

The reference calculation uses Python 3.12, NumPy 2.0.2, python-flint 0.8.0, and Arb precision 192 bits. The floating midpoint inverses and KKT witnesses are frozen point preconditioners only. Every claimed interval bound and gate decision is computed by outward-rounded Arb arithmetic.

The reference phase vector θ_{ref} is declared in the theorem-bearing v0.7.4 source and is therefore bound by the source SHA-256 reported in Table 1; its fourteen values are reproduced in Appendix A so that the analytic statement can be read from the paper alone. It is not a separate backend-input artifact. Changing θ_{ref} changes the frozen source hash and defines a different computational statement. The binding is enforced at run time: the v0.9.3 verifier does not restate the model but injects its stage into the frozen v0.7.4 source, whose SHA-256 it checks against the value recorded in Table 1 before executing.

The source and reproduction instructions are available at `https://github.com/papasop/Geometric-Flow`. The exact frozen software release is archived in Ref. [15]; the exact release tag or commit should be reported when the artifact is cited.

Every constant that this paper quotes from the frozen certificates and uses in a theorem or proof inequality is a *direction-safe* decimal: lower bounds are truncated downwards, positive upper bounds are relaxed upwards, and negative upper bounds are relaxed towards zero. Quantities the paper then derives from these, such as the complete-interval decrease of Corollary 2, are rendered by the same convention from the direction-safe inputs. The diagnostic values of Appendix B.1 are not theorem-bearing and are quoted exactly as recorded. The corresponding certificate fields are stored as binary64 renderings of the underlying Arb enclosures and are therefore rounded to nearest, not outwards; they should not be read as exact decimal bounds. Table 2 gives the correspondence for the theorem-bearing and proof-bearing constants.

Table 2: Direction-safe decimals quoted in the text and the binary64-rendered certificate fields from which they are derived.

Quantity	Quoted (direction-safe)	Certificate field	binary64 rendering
$\|\mathrm{grad}_{\mathcal{F}_3} L_6\|$ lower	0.64195291915915	`intrinsic_projected_gradient_norm_lower`	`0.6419529191591549`
dL_6/dt upper	-0.64195291915915	`uniform_dL6_dt_upper`	`-0.6419529191591549`
Picard contraction factor	0.0034878591567594	`picard_contraction_factor`	`0.00348785915675938`
Picard self-mapping utilization	0.00058130985945990	`picard_self_mapping_utilization`	`0.0005813098594598967`
Cauchy–Lipschitz bound	$3.4878591567594 \times 10^{11}$	`cauchy_lipschitz_upper`	`348785915675.93805`
graph (Krawczyk) utilization	0.023161050514437	`complex_graph_krawczyk_utilization`	`0.02316105051443677`
normal-derivative defect	0.00056424395018765	`implicit_derivative_neumann_defect_upper` and `complex_graph_normal_derivative_defect_upper` (same expression, same value)	`0.0005642439501876456`
pullback-metric defect	0.00069535968103773	`pullback_metric_neumann_defect_upper`	`0.0006953596810377727`
response-tangency norm upper	$2.3071147819355 \times 10^{-9}$	`maximum_response_tangency_norm_upper`	`2.3071147819354663e-09`
$\|P\nabla L_6\|$ lower	0.65307847481072	`minimum_projected_gradient_norm_lower`	`0.6530784748107296`
oriented pairing upper	-0.65307847481365	`tangent_projected_gradient_upper`	`-0.6530784748136524`
$dL_6/d\ell$ upper	-0.65307846977005	`maximum_dL6_dell_upper`	`-0.6530784697700559`
right-inverse defect upper	0.12993443884002	`maximum_right_inverse_defect_upper`	`0.1299344388400178`
curve-speed square enclosure	[0.999999999989079, 1.00000000000896]	`curve_speed_square_lower` / `_upper`	`0.9999999999890795` `1.000000000008951`

13 Limitations and global-flow outlook

The scope of Theorems 1 and 2 is intentionally narrow.

1. The complete-parent-box result is a descent theorem, not an ODE theorem.
2. The validated ODE theorem covers only one time step of duration 10^{-14} near child 15.
3. The paper does not establish complete-child traversal, endpoint inclusion and chaining of adjacent microsteps, or continuation through all ten charts. Later repository milestones

study finite same-chart continuation, but they are outside the theorem and artifact boundary of this paper.

4. No connection between arbitrary response-equivalent controls is proved.

5. No global fibre, topology, or holonomy theorem is claimed.

6. The Euclidean metric in the fourteen phase coordinates is declared, not derived uniquely from physics.

7. No statement about physical hardware, open-system dynamics, or experimental quantum-control performance is made.

These exclusions prevent a local computer-assisted result from being misreported as a global geometric-flow theorem.

These missing steps define the following local-to-global hierarchy:

$$\begin{array}{ll} \text{global response-fibre flow} & \longleftarrow \text{conjectural} \\ \text{validated ten-chart continuation} & \longleftarrow \text{open} \\ \text{validated complete-child ODE traversal} & \longleftarrow \text{open} \\ \text{validated intrinsic ODE microstep} & \longleftarrow \text{this paper} \\ \text{complete-parent-box strict descent} & \longleftarrow \text{this paper} \end{array}$$

14 Conclusion

We certified the existence and uniqueness of a local solution to the intrinsic normalized projected-gradient ODE on a six-dimensional local response fibre. Along the validated solution, $\mathcal{R}_3$ is exactly invariant and $dL_6/dt \leq -0.64195291915915$. A complementary parent-box certificate establishes the surrounding broad-domain descent geometry, with certified rate

$$\frac{d}{d\ell} L_6(\widehat{\gamma}(\ell)) \leq -0.65307846977005$$

in the affine atlas arclength coordinate ℓ, not in the Chebyshev coordinate s.

Thus exact invariance of a declared computational response need not imply dynamical stationarity of its implementation parameters. The main theorem establishes one local certified instance; complete-atlas and global continuation remain open.

Data and code availability

The theorem-bearing boundary of the present paper is frozen at the v0.9.3 release. Later commits in the same repository concern finite continuation and next-chart development and do not alter the claims proved here. The hash-bound source, inputs, protocol, certificate, report, and verification script are archived in the v0.9.3 GitHub release: `https://github.com/papasop/Geometric-Flow/releases/tag/v0.9.3-validated-ode-microstep`. No cloud service or QPU is required.

Conflict of interest

The author declares no conflict of interest.

A Frozen model constants

The reference phase vector of Section 2.2 is declared in the theorem-bearing v0.7.4 source, whose SHA-256 is recorded in Table 1. Its fourteen entries are listed below in the frozen source order, component j carrying the phase $\theta_{\mathrm{ref},j}$ of the j-th applied segment, so that $j = 1$ is the first segment applied to $|0\rangle$ and $j = 14$ the last, matching the propagator ordering of Eq. (6). The decimal literals are transcribed verbatim from the frozen source and are not reformatted through binary floating point.

j	$\theta_{\mathrm{ref},j}$	j	$\theta_{\mathrm{ref},j}$
1	`3.006797722681818`	8	`3.089644890790571`
2	`2.7106859720155914`	9	`-0.8755338801622679`
3	`1.1306621045783265`	10	`-2.6500043472817922`
4	`-2.6568476957176808`	11	`0.9588777193059705`
5	`1.4365241820035193`	12	`-3.1075630669100938`
6	`-2.0773016506803064`	13	`0.7072945305932086`
7	`0.16320548211467623`	14	`-0.48362649203822405`

With $|t\rangle = (t_1, t_2)^T$ the normalized nominal target determined by θ_{ref}, the algebraic orthogonal complement used in Eq. (7) is

$$|t^{\perp}\rangle = \left(-\overline{t_2},\ \overline{t_1}\right)^T.$$

Under $|t\rangle \mapsto e^{i\phi}|t\rangle$ this gives $|t^{\perp}\rangle \mapsto e^{-i\phi}|t^{\perp}\rangle$, so the phase-transformation law of Remark 1 can be checked directly from the paper.

These constants, together with the segment duration $\tau = 0.62$ and the nominal amplitude $\Omega = 1$ of Section 2.1, reproduce the source-bound model to which the certificates apply. Changing any of them changes the frozen source SHA-256 of Table 1 and defines a different computational statement.

B Computational diagnostics

The diagnostics recorded here explain the development history of the certificates. They do not enter the proof of Theorem 1.

B.1 The unresolved atlas-alignment diagnostic

For diagnostic alignment, the midpoint least-squares problem freezes $\lambda \in \mathbb{R}$ and $\mu \in \mathbb{R}^8$ in

$$w = \nabla L_6 + \lambda \frac{d\widehat{\gamma}}{d\ell} - J^T \mu. \tag{37}$$

Since $PJ^T = 0$, this gives $Pw = P\nabla L_6 + \lambda P\, d\widehat{\gamma}/d\ell$. The witness is formed with the same arclength derivative as the theorem-bearing gates. The proof encloses $\|w\|$ and adds an outward-rounded correction for the response-normal component of $d\widehat{\gamma}/d\ell$, yielding a rigorous but coarse parallel-residual bound. The witness calculation returns $\lambda_{\min} = 0.6530911830790997$, $\rho_{\|}^{\mathrm{upper}} = 0.008935710125297152$, and $\cos\alpha_{\min} = 0.9999600757453052$: the serialized curve is formally confined to a narrow cone around the projected-gradient direction under this enclosure, but the predeclared alignment gate $\rho_{\|} < 2 \times 10^{-4}$ is not satisfied. The complete-box audit is therefore fail-closed on its own namespace: the shipped v0.7.4 certificate records `all_gates_pass` = `false` and `validated_ODE_claimed` = `false`, whereas the independent intrinsic-chart certificate reports both flags `true` on its declared microscopic domain. That aggregate flag spans both audit stages. The evidence for Theorem 2 is the Stage-A namespace, in which the same certificate records `stage_a_rank_descent_cover_certified` = `true` and `child_boxes_passing_stage_a` = 16; the failing gate belongs to the Stage-B alignment diagnostic (`kkt_witness_alignment_cover_certified` = `false`).

The failed frozen-witness gate concerns identification of the serialized Chebyshev atlas with a projected-gradient trajectory. It does not contradict the separately constructed intrinsic fibre-chart ODE, whose existence and uniqueness are certified directly by a Picard argument. The gate remains a diagnostic about the old atlas only: it explains why the serialized curve is not certified as an ODE trajectory, and it is not a hypothesis of Theorem 1.

B.2 Why ambient tubes fail

Earlier development versions attempted to validate the projected-gradient ODE directly in an ambient Euclidean tube around the serialized atlas. That approach failed for structural reasons: a fourteen-dimensional Euclidean tube mixes the six fibre-tangent directions with the eight response-normal directions, so the enclosure couples motions that the constraint geometry keeps separate; the direct Gram-matrix inversion in ambient coordinates then suffers severe interval wrapping, destroying the contraction estimates. The tangent–normal fibre coordinates of Section 5 restore verifiability by separating the two directions before any interval arithmetic is applied.

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